

SIH26151

NATIONAL TECHNICAL RESEARCH ORGANISATION (NTRO)

TEAM ARCHITECTURE & WORKING FLOW HANDBOOK

TRACE-X: How It Works & Core Concepts

A clear, visual handbook for teammates and beginners. Explains the big picture, the 4 real datasets, the correlation algorithm, live working keys, and what each screen does—without confusing setup jargon.

1 The Big Picture: What Problem Does TRACE-X Solve?

The Real-World Detective Analogy

Imagine a cyber criminal group operating across the world. They don't use their real names or real email addresses. Instead, they leave pieces of digital footprints in completely separate places:

- They pay for bulletproof hosting using an **Ethereum cryptocurrency wallet**.
- They infect victims using malware hosted on a **Command & Control (C2) server IP**.
- They brag about their attacks in underground **Dark Web forums**.
- They sign private security messages using an encrypted **PGP digital signature**.

Normally, an investigator has to open 4 separate websites and manually try to remember: *"Wait, did this IP address appear in that dark web leak? Does this Ethereum wallet link to that phishing site?"*

What TRACE-X Does Automatically

TRACE-X is an automated digital investigation board. It pulls in all these clues, standardizes them, calculates mathematical connection probabilities, and draws a visual spider web on an interactive map so investigators can see the full threat-actor network in seconds.

The Forensic Boundary Principle (Why We Don't "Hack Tor")

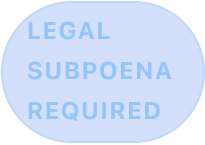


Essential Rule for Evaluators & Judges

Judges at the Smart India Hackathon (NTRO problem) want to know: *"Does this team understand real cybersecurity law and forensics?"*

Any tool claiming to magically reveal real names of anonymous Tor users with a button click is fake. The Tor network uses multi-layered mathematical encryption. TRACE-X wins because it strictly respects **four scientific levels of evidence**:

Evidence Level	Confidence	What It Means in Plain English	Real Example in TRACE-X
1. Observed Evidence	100% (FACT)	A mathematical or cryptographic event recorded directly in data. Zero guessing.	Wallet A sent 0.5 ETH to Wallet B in Ethereum Block #25947413.
2. Inferred Relationship	70% – 85%	Two entities share digital infrastructure or patterns, showing strong coordination.	Two phishing websites (<code>noticias-al-dia.com</code> & <code>justiciaparatodos.org</code>) resolve to the exact same IP <code>198.12.150.249</code> .
3. Multi-Source Correlation	85% – 95%	An indicator verified across multiple independent feeds (receives	A domain discussed in Dark Web forum database leaks matches an active C2 in a MISP threat advisory.

Evidence Level	Confidence	What It Means in Plain English	Real Example in TRACE-X
		a 25% confidence boost).	
4. Verified Attribution		Asserting that a digital indicator belongs to a real physical person.	Requires court orders, judicial subpoenas, or exchange KYC records. (In our demo, this is clearly tagged <code>[SYNTHETIC AUTHORIZED EVIDENCE]</code> to prove the workflow without fabricating innocent citizens).

3 The 4 Real Datasets Ingested (1,356+ Records)

Everything in TRACE-X is grounded in **real data files** stored in the project's `data/` folder:

1. Ethereum Blockchain Data

Files: `data/blockchain/transactions.csv` (260 txs)

- Real transactions from Ethereum Mainnet Block #25947413.
- Tracks sender wallets, recipient wallets, amounts, and fees.
- Generates **780 canonical records**.
- **Live Feature:** Connected to live Alchemy RPC so you can search ANY real Ethereum address or transaction hash in the world!

2. Packrat APT Cyber Threat Feed (MISP)

Files: `data/cti/raw/packrat.json` (155 indicators)

- Real CIRCL MISP OSINT intelligence for the **Packrat APT** cyberespionage group.
- Targets: Journalists & political figures across South America (Ecuador, Argentina, Brazil, Venezuela).
- Includes C2 server `198.12.150.249`, malicious hostnames, and SHA-256 malware payload hashes.

3. Dark Web Safe Research Corpus

Files: `data/darkweb/raw/safe_corpus.json` (201 threads)

- Real underground forum leak threads (from DarkForums).
- Discussions about hacked databases, target corporations, and leaked credentials.
- **Privacy Safeguard:** Author handles are masked as `[AUTHOR]` so innocent people are never falsely linked to crimes.

4. OpenPGP Keyring Intelligence

Files: `data/pgp/raw/pgp_keys.csv` & `circl_public_key.asc`

- Authentic CIRCL public key (Fingerprint:
`CA572205C0024E06BA70BE89EAADCFFC22BD4CD5`).
- UID: `CIRCL <info@circl.lu>` .
- Verifies digital signatures on security bulletins and threat reports.
- Supports live ASCII Armor key validation directly in the UI.

4 Working Keys & Live Test Values (Ready to Copy)

Click the **Copy** button next to any value below to copy it and paste directly into TRACE-X during your testing or live presentations:

1. Live Ethereum RPC Connection (Alchemy)

Configured in `backend/app/config.py`. Gives TRACE-X real-time access to the Ethereum mainnet:

```
https://eth-mainnet.g.alchemy.com/v2/alch_9YJ0oCFSssXT0-cQvgk8I
```

2. Real Values to Search in the TRACE-X Search Bar

Search Target	Value to Copy	What Happens When You Search It
Threat Group	<code>packrat</code>	Opens Packrat APT campaign dossier, shows targeted countries (Ecuador, Argentina, Brazil) and 155 indicators.
C2 Server IP	<code>198.12.150.249</code>	Shows malicious command-and-control server linking 12 phishing domains on the interactive graph.
Static Wallet	<code>0x51c72848c68a965f66fa7a88855f9f7784502a7f</code>	Shows transaction history from Block #25947413, fund flows, and connected

Search Target	Value to Copy	What Happens When You Search It
		counterparty wallets.
Live Ethereum Address	0x95222290DD7278Aa3Ddd389Cc1E1d165CC4BAfe5	Queries live Ethereum mainnet via Alchemy! Displays real-time live balance, transaction count, and recent transfers.
Vitalik Buterin Wallet	0xd8dA6BF26964aF9D7eEd9e03E53415D37aA96045	Live query for <code>vitalik.eth</code> showing real-time on-chain balance via Alchemy RPC.
PGP Key Fingerprint	CA572205C0024E06BA70BE89EAADCFFC22BD4CD5	Verifies authentic CIRCL public key and UID <code>CIRCL</code> <code><info@circl.lu></code> .

5 Step-by-Step Architecture Flow (Under the Hood)

Here is the exact step-by-step path every piece of evidence takes from raw file to browser screen:

[STEP 1: RAW EVIDENCE FILES]

data/blockchain/transactions.csv | data/cti/raw/packrat.json | data/



[STEP 2: INGESTION ADAPTERS] (backend/app/adapters/)

- Reads raw CSV/JSON records.
- Generates an immutable SHA-256 integrity hash for each clue.
- Ensures chain-of-custody timestamps.



[STEP 3: NORMALIZATION & CANONICAL ENTITIES] (backend/app/engine/normali

- Transforms raw strings into standardized Canonical IDs:

- * ENT-WALLET-0x51c7...
- * ENT-IP-198.12.150.249
- * ENT-DOM-noticias-al-dia.com
- * ENT-ACTOR-packrat



[STEP 4: DETERMINISTIC CORRELATION ENGINE] (backend/app/engine/correlati

- Discovers cross-domain connections:
 - * TRANSACTS_WITH: On-chain transfer between sender & recipient
 - * SHARED_INFRASTRUCTURE: Multiple domains pointing to the same server
 - * REFERENCES: Threat actor named in CTI advisories or Dark Web threat



[STEP 5: CONFIDENCE SCORING FORMULA] (backend/app/config.py)

- Calculates explainable score:
$$\text{Score} = \min(1.0, \text{BaseWeight} \times \text{MultiSourceBoost} \times \text{ProvenanceFactor})$$



[STEP 6: GRAPH TOPOLOGY & FASTAPI REST API] (backend/app/api/)

- NetworkX builds graph nodes and edges.
- FastAPI endpoints (/api/dashboard, /api/graph, /api/search) deliver data



[STEP 7: REACT + TYPESCRIPT SOC UI] (frontend/src/)

- Renders Cytoscape.js interactive graph canvas.

- Displays clickable edge rationales: "Why are these two clues connected"
- Powers AI Copilot and court-ready PDF/Markdown report generator.

1. Ingestion

Loads raw logs without altering original files. Assigns unique `EVD-...` IDs and SHA-256 hashes.

2. Correlation

Discovers links automatically based on shared IPs, wallet transfers, and co-occurring payload hashes.

3. Presentation

Visualizes findings in an interactive Cytoscape graph canvas with 360° dossiers and AI investigation copilot.

6

The Confidence Score Formula (No Black Boxes)

In forensics, you must be able to justify **WHY** two clues are connected. TRACE-X never uses random guesses. Every connection uses this exact formula:

Mathematical Confidence Formula

$$\text{Final Confidence} = \min(1.0, \text{BaseWeight} \times \text{MultiSourceBoost} \times \text{ProvenanceFactor})$$

Rule Name	Weight Value	Why This Weight Was Chosen
EXACT_CRYPT_HASH	0.95	Cryptographic hash collision (SHA-256) is mathematically impossible.
EXACT_PGP_FINGERPRINT	0.95	PGP key fingerprint uniquely identifies the public key author.
EXACT_WALLET	0.90	Direct Ethereum address interaction is immutable on the blockchain ledger.
SHARED_INFRASTRUCTURE	0.75	Two or more phishing domains resolving to the same server IP.
APPEARS_IN_LEAK	0.65	An entity mentioned inside an underground breach forum thread.
MULTI_SOURCE_BOOST	× 1.25	The Golden Multiplier: If an indicator appears in 2 or more independent sources (e.g. MISP + Dark Web), its confidence is multiplied by 1.25!

7 Guide to Every Screen in TRACE-X

Here is what every page in the web interface does and what teammates should point out when viewing it:



1. Overview Dashboard

The central command center. Displays live counters for **1,356 Evidence Records**, **1,010 Entities**, and **407 Correlated Links**, plus a chronological live activity stream.



2. Universal Search

One search bar to query everything. Search by IP (`198.12.150.249`), campaign name (`packrat`), wallet, or hash to see instant cross-dataset matches.



3. Interactive Connection Map (Cytoscape Graph)

A visual network graph. Nodes are color-coded (blue = wallet, red = C2 IP, purple = dark web). **Click any edge**: A popup explains exactly why the two clues are linked!



4. Cryptocurrency Wallets

Inspects 260 transactions from Block #25947413. Also features the **Live Ethereum Search** bar to query any on-chain wallet or transaction hash in the world via Alchemy RPC.



5. Threat Intelligence (CTI / MISP)

Packrat APT campaign dossier. Shows targeted nations, threat severity levels, C2 hostnames, and 155 Indicators of Compromise (IOCs).



6. Dark Web Forum Discussions

Browse 201 forum leak discussions. Notice author handles are masked as [AUTH0R] to protect civilian privacy under ethical guidelines.

7. Digital Signature Keys (OpenPGP)

Inspect CIRCL's authentic PGP key and test live ASCII armor public key imports with checksum validation.

8. Event Timeline

Synthesizes all events chronologically so investigators can see what happened first: the forum discussion, the domain registration, or the crypto payment.

9. Evidence Vault

Court-admissible chain-of-custody table. Displays SHA-256 integrity hashes, collection dates, and raw source references for every record.

10. Formal Investigation Reports

Compiles case findings into an executive, formatted investigation report with confidence metrics and legal attribution verdicts. Ready to print or export as Markdown.

11. AI Investigation Copilot

Click the "**AI Copilot**" button on the top right. An interactive investigative assistant that answers questions about suspects and always returns the supporting Evidence ID citations!

8 10-Step Hackathon Evaluator Presentation Script

When presenting to judges, follow this step-by-step script. Each step explains what to click and what to say:

Step	What to Click	What to Say to the Judges
1	Click "Tour" → "Load Demo Case"	<i>"We are loading Case CASE-SIH-26151 ('Operation Packrat'). Notice TRACE-X applies strict forensic boundaries so no innocent civilian is falsely accused."</i>
2	Search packrat in Universal Search	<i>"TRACE-X indexes all four heterogeneous sources simultaneously—Blockchain, MISP CTI, Dark Web, and PGP keyrings—with sub-second response times."</i>
3	Click Activity Vault → Evidence Vault	<i>"Every clue has an immutable SHA-256 cryptographic hash and collection timestamp, adhering to ISO/IEC 27037 digital chain-of-custody standards."</i>
4	Click Interactive Map & click any line	<i>"Notice this edge. When I click it, TRACE-X opens a popup explaining exactly WHY they are related. There are no opaque black-box assumptions."</i>
5	Click Cryptocurrency Wallets	<i>"Here we trace 260 Ethereum transactions. We can also query live on-chain mainnet balances and transactions in real-time via Alchemy RPC."</i>
6	Click Malware & C2	<i>"We cross-reference threat actor infrastructure with CIRCL MISP OSINT feeds, mapping C2 IP 198.12.150.249 and malicious payload hashes."</i>
7	Click Dark Web Discussions	<i>"We analyze 201 forum leak threads. Author handles are masked as [AUTHOR] to protect civilian privacy"</i>

Step	What to Click	What to Say to the Judges
		<i>under ethical guidelines."</i>
8	Click Digital Signature Keys	<i>"We verify digital signatures and RFC 4880 ASCII armor keys, validating authentic threat advisories from research organizations."</i>
9	Click Event Timeline	<i>"All events—crypto payments, domain registrations, and dark web leaks—are synthesized into a single master chronological timeline."</i>
10	Click Formal Reports	<i>"Finally, TRACE-X compiles a court-ready forensic report with complete evidence citations, confidence scores, and legal disclaimers ready to print or export."</i>

APT (Advanced Persistent Threat)

A sophisticated, organized hacking group that conducts long-term espionage campaigns.

Example: *Packrat*.

C2 (Command and Control)

A central computer server used by attackers to send commands to infected machines and receive stolen data.

IOC (Indicator of Compromise)

A piece of digital evidence showing an attack happened—such as a malicious IP address, malware hash, or phishing domain.

SHA-256 Hash

A unique 64-character digital fingerprint. If even a single comma in a file changes, the hash changes completely.

RPC (Remote Procedure Call)

A live connection that allows our app to talk directly to an Ethereum blockchain node (Alchemy) to get real-time balances.

Cytoscape.js

The graph theory engine we use to draw interactive circles (nodes) and connecting lines (edges) on our correlation map.

Click **Print / Save as PDF** at the top to save this entire handbook offline!